H10 versus D2
Sleep-staging heuristics

10-page comparison with peer-reviewed evidence
Implemented H10/OpenBeat rules versus reverse-engineered D2 patterns | 6 November 2026

Recommendation: improve the H10 classifier through signal-quality handling, probabilistic stage decisions and validated temporal modelling. Use the inferred D2 patterns as hypotheses to test, not as manufacturer rules or ground-truth sleep stages.

Three different kinds of evidence

Evidence layer	What is known
H10 + OpenBeat	Exact code in supplied src/sleep.ts, method 2: thresholds, coefficients, filtering and smoothing. This is OpenBeat's algorithm, not a built-in Polar H10 sleep classifier.
Reverse-engineered D2	Observed stage labels, associated parameters, bout durations and held-out surrogate tests from the D2 export. Exact Huawei firmware logic remains unknown.
Published research	PSG-referenced studies of cardiac/respiratory/PPG staging, Huawei GT2 validation and a D2 blood-pressure study. These are separate evidence streams.

The practical conclusion

The current H10 heuristic is transparent and uses useful physiological signals, but its coefficient values, fixed REM gate and stage-bout rules have no demonstrated PSG calibration in the supplied material. D2 data show substantial overlap in the exported stage parameters, so replacing these rules with a few inferred D2 cutoffs would not solve the problem.

Across the two non-demo paired nights, H10/OpenBeat and D2 match on 44.9% of specific-stage minutes, with pooled kappa -0.013. This establishes disagreement, not which system is accurate. The four eligible paired recordings and the D2-only reverse-engineering results are reused here; no new PSG labels or improved-classifier performance are claimed.

This is a targeted literature-informed technical assessment, not a systematic review. Sources were checked on 6 November 2026. All demo data remain excluded. References and local provenance are on page 10.

01 Awake and light: direct comparison

Implemented H10 criteria versus the D2 criteria that could actually be inferred.

Stage	H10/OpenBeat method 2	Reverse-engineered D2 evidence
Awake	Wake candidate if activity >max(8 mG, 2.5 x median activity), chest turn >20 degrees, or HR >20th-percentile baseline +12 bpm with activity >half the movement threshold. At least 2 of 3 epochs must support wake.	HR >=55 bpm is the strongest simple association: binary balanced accuracy 73.6% against D2 awake labels, but precision only 21.8% among observed-HR minutes. It is not a confirmed firmware threshold.
Light	Default sleep label when neither deep nor REM rule wins; also absorbs short deep/REM bouts during smoothing.	Code 1 is light. HR median 52 bpm and associated HRV median 47 ms overlap the other stages. No reliable distinct light threshold was recovered.
Sleep onset	Backdated to the first 10-minute continuous quiet/data-present run. End determined from the last qualifying 5-minute run.	Exported onset and boundaries are available, but the hidden detection thresholds are not. Awake-to-sleep transitions usually enter light.

What to change in the awake heuristic

The D2 result supports checking relative cardiac activation, but not copying 55 bpm into H10 code. A fixed personal-sample threshold is incompatible with different resting rates, sensor processing and observation windows. Moreover, when boundary wake minutes are removed, its precision falls to 10.5% for internal wake.

The H10 rule requires movement for its elevated-HR path and can therefore miss quiet wake. Retain cardiac and movement evidence as complementary inputs, add a confidence output, and test quiet-wake performance against PSG rather than assuming immobility is sleep. Treat diary-reported quiet wake as a useful audit aid, not a definitive training label.

What to change in light classification

A failed deep/REM test and insufficient signal quality are different situations. Preserve light as a physiological hypothesis; add an explicit uncertain/unstaged output when inputs cannot support a specific decision. Otherwise missing or ambiguous data are systematically turned into apparent light sleep.

Sources: local sleep.ts and D2-only reconstruction (658 observed-HR minutes; 3,791 rows in the physiological-model cohort). Clinical validation principles: [8].

02 Deep and REM: exact rules and evidence

The D2-derived parameters do not supply a replacement lookup table.

Implemented H10 scores

Deep = -zHR - zCV + 0.5 zRMSSD - 0.5 zHRspread + 0.5(1-2t)
REM = zHR + zHRspread + zCV - 0.3 zRMSSD + 0.5(2t-1)

Deep wins if its score >1.2. Otherwise REM requires score >1.2 and elapsed time after onset >50 minutes; otherwise light. For sleep periods <3 hours, t=0.5; otherwise t is relative progress through the detected period.

HR, RMSSD and HR spread use local relative normalization; respiratory CV combines standardized chest-derived and RR-derived irregularity. Missing normalized terms become zero. Deep has decision-order priority when both scores would qualify. These scores are not calibrated probabilities.

Stage	D2-derived observations	Implication for H10
Deep	Median HR 51 bpm; associated HRV 45 ms. HR <=54.5 rule: held-out binary BA 54.2%; HRV <=39.5: 50.4%.	Low-HR regularity remains a plausible feature, not a sufficient N3 criterion. D2 does not justify the H10 coefficient signs or threshold 1.2.
REM	Median HR 49 bpm; associated HRV 48 ms. HR <=53.5: held-out BA 58.0%; HRV >67.5: 51.7%.	The D2 export does not support a universal higher-HR REM rule or a reliable HRV cutoff. Use multivariate, time-dependent evidence.

Why an apparent contradiction is not a new physiological law

D2-labelled REM has lower pooled observed HR than D2-labelled deep. That does not prove H10's positive REM/HR coefficient should be reversed. Exported readings are sparse, feature windows differ, nights have different baselines and other internal inputs are missing. Fit conditional relationships against reference labels rather than changing a sign from these pooled medians.

A better transparent model would estimate separate stage likelihoods and compare all candidates jointly. Evaluate deep-versus-REM conflicts explicitly instead of resolving them by code order. Recalibrate thresholds and probabilities on independent PSG-labelled data; a softmax applied to current untrained scores alone would not establish valid probabilities.

Sources: supplied sleep.ts: localZ, analyzeSleep; prior D2-only threshold tests. BA here is one target versus all other specific stages, with 50% as the non-discriminating reference.

03 Timing, naps and signal quality

This is where the inferred D2 behavior most clearly challenges the current H10 design.

Mechanism	H10 implementation	D2 observed behavior / limitation
Short stage bouts	Seven-epoch mode filter plus a nominal five-minute minimum-bout pass; short deep/REM runs can become light.	No observed light/deep/REM bout shorter than 3 minutes across 94 records. This suggests smoothing or reporting compression, not proof of a hard 3-minute firmware rule.
Early REM	Direct REM eligibility blocked through the first 50 minutes after estimated onset.	First exported REM is as early as 7 minutes; 34/57 specifically staged records show REM within 50 minutes. Record truncation can affect apparent latency.
Naps	Short periods lose the early/late prior, but still face the REM gate and smoothing rules.	37 generic nap records use code 5: asleep without a stage. They cannot validate H10 nap-stage labels.
Missing inputs	An epoch has data if activity OR HR exists. Missing standardized feature terms are zero.	Exported inputs are too sparse to recover Huawei's internal quality gates. Absence of exported steps is not stillness.

Recommended timing changes - to be tested, not assumed better

Replace the hard REM exclusion with a soft, context-dependent prior and allow contrary signal evidence. Avoid forcing every deep/REM episode to five minutes: compare a shorter or learned duration penalty against PSG, including naps and fragmented sleep. Do not simply copy the D2's observed three-minute minimum.

Separate a recording's start from true physiological sleep onset. Keep uncertainty when recording begins after sleep has started. The app's local-clock nap label (midpoint between 08:00 and 20:00) should not stand in for physiology, especially for shift workers; the stage rules themselves should use documented context.

Recommended quality changes

In the paired night reports, H10 breathing estimates are missing in 20.5% and 12.7% of stored epochs, despite reported coverage of 100%. Track valid RR, movement and respiration separately. Let confidence fall when discriminating features disappear rather than treating missing evidence as an average feature value.

Evidence: stored paired feature arrays and D2-only sequence analysis. Smoothing can alter final labels near the 50-minute boundary; the code's gate applies to direct REM decisions before smoothing.

04 Peer-reviewed cardiac staging research

Relevant precedents support richer features and temporal learning, not magic HR cutoffs.

Study / task	Inputs and validation	Reported performance
Fonseca et al., 2020 [1] Four classes	132 ECG-derived HRV features + body movement; independent hold-out cohort of 389 sleep-clinic recordings.	Accuracy 75.9%; mean kappa 0.60.
Sridhar et al., 2020 [2] Four classes	Instantaneous HR sequence; >10,000 training/validation nights across SHHS/MESA; additional external cohort of 993 subjects.	Held-out SHHS accuracy 77%; kappa 0.66. These are not stated as the external-cohort metrics.
Sun et al., 2020 [3] Five and three classes	8,682 PSGs; neural temporal models using ECG timing and respiratory effort. Best combination used ECG + abdominal effort.	Kappa 0.585 for five stages; 0.760 for wake/REM/NREM. Different task from four-class staging.
Radha et al., 2021 [4] Four classes, PPG	ECG pretraining and adaptation to wrist PPG; PPG data from 60 healthy participants, 101 recordings.	Accuracy 76.36 +/-7.57%; kappa 0.65 +/-0.11. Not a D2 study.

What can reasonably be transferred to H10

These studies show that cardiac signals contain useful sleep information when models use richer feature structure and temporal context. They do not validate OpenBeat's two linear scores, hand-selected coefficients or minimum-bout rules. Their results are benchmarks for study design and model development, not expected performance for this app.

A practical first step is a small interpretable multivariate classifier with a temporal layer, using calibrated cardiac, movement and respiration features. A large neural network is not the only option and is inappropriate to train from two paired nights. Public PSG ECG data can support initial development, but chest acceleration and a real H10 recording environment require additional validation.

Important comparisons to avoid

Do not rank the studies by raw accuracy across different classes, populations and datasets. Do not equate ECG+respiratory-belt results with chest-accelerometer respiration. Do not assume an ECG-trained model works unchanged on wrist pulse timing: the PPG transfer study specifically addressed a sensor/domain change. [3,4]

All four entries are peer-reviewed original studies. Rounded reported metrics preserve their original task definitions; no confidence interval is invented for the user's app.

05 Huawei evidence and the CPC distinction

Research relevant to D2 does not establish its hidden stage thresholds.

Huawei GT2 PSG validation: informative, but a different watch

Guo et al. studied 98 sleep-clinic participants with simultaneous Huawei GT2 and PSG. Table 2 reports one-versus-rest mean kappa of 0.12 for light, 0.32 for deep and 0.12 for REM, compared with 0.43 for binary sleep detection. Table 3 reports total sleep overestimation of 28.47 minutes and WASO underestimation of 37.00 minutes. These are GT2 results, not D2 validation. [5]

Evidence	What it supports	What it does not support
D2 study: Liu et al., 2025 [6]	85 participants for blood-pressure validation; 46 for nocturnal comparison and sleep impact. Describes ACC/PPG TruSleep and notes then-unpublished internal PSG validation.	No published D2 four-stage confusion matrix in this study. BP accuracy is not proof of sleep-stage accuracy.
Huawei family documentation [9]	Uses motion and physiological features; earlier generations described heart-respiration coupling.	Does not disclose exact D2 weights or prove which TruSleep version ran on this watch.
Original CPC paper: Thomas et al., 2005 [7]	Single-lead ECG coupling analysis captures stable/unstable sleep physiology and relates more closely to CAP patterns than standard stages.	High coupling is not a one-to-one N3 label. Stable NREM must not be renamed deep sleep automatically.

Why this matters for H10 feature development

OpenBeat averages two respiratory-irregularity z-scores; that is not cardiopulmonary coupling analysis. A coupling feature would measure how cardiac interval oscillations and respiration interact across frequencies/time. It is a promising candidate input, but its implementation and mapping to stage labels need separate testing. [7]

Add a distinct cardiorespiratory-stability output if useful, while retaining deep/REM as uncertain stage predictions. This avoids turning a physiologically meaningful stability measure into an unjustified claim of EEG-defined N3 sleep.

A targeted search located D2 BP/sleep-impact evidence but did not establish dedicated public D2 four-stage PSG validation. This is a search limitation, not a claim that no such study can exist.

06 First changes to the H10 heuristic

Prioritize information integrity and testable decisions before adding complexity.

Priority / source area	Suggested improvement	Why / acceptance check
P1: hasData, coverage and features	Store per-epoch validity for RR, HRV, acceleration and respiration, plus feature-window age and rejection rate.	100% activity coverage must not imply complete cardiac/respiratory evidence. Verify missing-data stress tests.
P1: missing standardized terms	Keep missingness explicit; use a validated fallback or abstain from specific staging. Do not silently make every missing feature zero.	An absent measure and an average standardized measure are not equivalent. Evaluate confidence and coverage together.
P2: rawWake / onset	Combine movement and relative cardiac evidence with probabilities; audit quiet wake and recording-boundary onset.	Optimize wake precision and recall against PSG, including wake without large movement.
P2: deep/REM score arbitration	Compare stage likelihoods jointly; calibrate with held-out reference labels. Remove untested deep-first priority as a default decision policy.	Reduce arbitrary conflicts; report calibrated probabilities and a confusion matrix.
P2: timing / minBouts	Test soft onset/REM priors and duration-aware smoothing in place of hard 50-minute and 5-minute rules.	Preserve genuine short bouts while controlling flicker. Validate naps separately.

Normalization and feature reliability

Evaluate robust local scaling, such as median/IQR or median/MAD with a scale floor, alongside the current mean/SD residual scaling. Use it as a candidate, not a guaranteed improvement: short windows, artefacts and low variability can destabilize either method. Keep normalization parameters reproducible and avoid using the unknown target stage to define baselines.

The current local window reaches into the future by up to 45 minutes. That is acceptable for retrospective analysis if disclosed; it cannot be used unchanged for a real-time claim. Benchmark a causal version separately. Likewise, do not hide rejected irregular beats: distinguish unreliable RR from genuine rhythm irregularity, and lower stage confidence where necessary.

These are proposed engineering experiments, not measured accuracy gains. Relevant code: src/sleep.ts: heartFeatures, localZ, analyzeSleep, modeFilter and minBouts. No application files were changed for this report.

07 A stronger, still transparent design

Use D2-derived observations to prioritize hypotheses, and PSG labels to fit the model.

Feature set: useful additions, not unrestricted feature collection

Retain accepted-RR rate, HR trend, RMSSD and movement. Test short- and longer-window HRV, Poincare/nonlinear regularity measures, breathing rate/irregularity, posture-change history and cardiac-respiratory coherence. Require sufficient valid data for each calculation. Avoid treating LF/HF or one entropy feature as a unique REM/deep marker.

Chest-derived breathing and RR respiratory modulation are related estimates, not two independent votes. Test quality-weighted combination and disagreement flags. If ECG waveform data are available and reliable, ECG-derived respiration can be evaluated as another signal; the current sleep classifier does not use ECG morphology. [3,7]

An interpretable probabilistic alternative

Fit a multinomial logistic model or small boosted/tree model to PSG-labelled epochs, with explicit missingness and a limited, prespecified feature set. Estimate wake, light, deep and REM likelihoods jointly. A hierarchical sleep/wake then sleep-stage model is another candidate; compare it directly rather than assuming it is superior.

Apply a hidden/semi-Markov temporal model with learned transition and duration tendencies. Use soft penalties, permit supported atypical transitions and retain short wake events. D2's three-minute observed bout floor and frequent awake-to-light transitions are candidate checks, not duration constraints to hard-code.

How the reverse-engineered D2 criteria should be used

Use HR >=55 only as an association to audit after personalization; its low precision argues against adoption as a wake rule. Use the poor D2 parameter separation as evidence to avoid single-value stage cutoffs. Use early exported REM to question the universality of the H10 gate, while remembering that truncated D2 records cannot establish true REM latency.

Training on D2 labels would create a D2 imitation model. It may be useful for compatibility experiments but must remain distinct from a PSG-validated sleep-stage model.

08 Validation and improvement roadmap

Measure whether each change improves staging, rather than just matching D2 more closely.

Experiment	Design and interpretation
Baseline	Freeze current method 2; preserve raw inputs, output probabilities if available, method version and processing settings.
Quality ablation	Compare current zero-substitution with validity-aware estimation / abstention. Report stage performance at matched accepted-epoch coverage.
Timing ablation	Individually test removal of the REM gate, shorter/learned bout penalties and a softer time-of-night prior. Avoid changing all simultaneously.
Feature ablation	Add cardiorespiratory coherence and expanded HRV in separate steps; compare against HR + movement and time-only baselines.
Model comparison	Current scores versus a small trained model versus the same model plus temporal decoding; keep test subjects completely unseen.
Domain / subgroup tests	Validate real H10 data, nights and naps, younger/older users, fragmented sleep and relevant sleep-disorder populations; retain firmware and algorithm versions.

Reference and metrics

Use simultaneous PSG scoring for the four-class target wake / N1+N2 / N3 / REM. Match epoch definitions and synchronize clocks; preserve uncertain alignment rather than selecting a test-set offset that maximizes agreement. Split training and testing by person, not by adjacent minutes. If using public ECG PSG for development, add a separate H10-domain test.

Report per-stage precision/recall/F1, macro F1, balanced accuracy, Cohen's kappa, calibration and total-sleep/WASO errors. Include confusion matrices, bout/transition errors and error by feature quality. Bootstrap at participant/night level, not at individual-epoch level, to respect dependence. Evaluate both accuracy and the amount of data on which the model abstains.

Decision rule for releasing a change

Choose acceptance criteria before testing: a reproducible improvement in the prespecified reference metrics without unacceptable loss of wake detection, minority-stage performance or usable coverage. A more plausible-looking hypnogram or higher H10-D2 agreement is insufficient. The roughly 76-77% four-class research results are examples of what trained models achieved on their datasets, not release targets guaranteed for OpenBeat. [1,2,4]

Current evidence comprises two non-demo paired nights, two naps and sparse D2-only parameters. It cannot calibrate a general H10 sleep classifier or demonstrate the benefit of these proposed changes.

09 References and conclusion

Peer-reviewed evidence, source-code provenance and the distinction that matters.

[1] Fonseca et al. Sleep. 2020;43(9):zsaa048. HRV + movement, recurrent model.

[2] Sridhar et al. npj Digital Medicine. 2020;3:106. Instantaneous-HR staging.

[3] Sun et al. Sleep. 2020;43(7):zsz306. ECG and respiration staging.

[4] Radha et al. npj Digital Medicine. 2021;4:135. ECG-to-PPG transfer learning.

[5] Guo et al. PLOS ONE. 2025;20(9):e0330774. Huawei GT2 versus PSG.

[6] Liu et al. Journal of Clinical Hypertension. 2025;27(11):e70183. D2 BP / sleep impact.

[7] Thomas et al. Sleep. 2005;28(9):1151-1161. ECG cardiopulmonary coupling.

[8] Schutte-Rodin et al. Journal of Clinical Sleep Medicine. 2021;17:2275-2282. AASM update.

[9] Huawei Support. Introduction to sleep monitoring principles. Manufacturer context.

Local evidence used

Backup: /Users/mike/Downloads/openbeat-backup-2026-11-06-10-57.json
App version 1.32.1 (5 Nov 2026); saved sleep method 2. H10 source: src/sleep.ts. D2 source: src/d2/extra.ts, sleepfile.ts and night.ts; synchronization: src/syncsleep.ts.
Source root: /Users/mike/Desktop/01 Elitebook 13 Apr 2026/Documents/programming/ECG Analysis/GitHub uploads/2026-11-06 0001 publish/

Paired analysis: 657 shared specifically staged overnight minutes, 295 matches, kappa -0.0129; no PSG reference. D2 reconstruction: 94 records, 23,221 specific-stage minutes; 658 observed-HR minutes; 3,791 physiological-model rows across 19 held-out date groups. HRV associations reuse the latest reading within five minutes and are not independent measurements. Demo data excluded.

What follows from the combined evidence

The most defensible H10 improvement is not to copy the inferred D2 thresholds. First make signal validity and uncertainty explicit, then replace hard timing and ordered stage decisions with trained, testable likelihoods and soft temporal structure. The D2 patterns identify useful questions; only independent PSG-labelled validation can establish whether an H10 change improves sleep staging.

Research search: targeted primary-source searches for ECG/HRV/respiration sleep staging, Huawei PSG validation and D2-specific evidence, accessed 6 November 2026. This report neither certifies either device nor claims that the proposed classifier changes have been implemented or validated.